

Statistics for Economists II

Course Introduction

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Aim of the Course

- The course is intended to provide the applied knowledge and skills to enable a student to analyse various data using various statistics tools, as well as give students solid foundation for intermediate statistical and econometric analyses

Course Objectives

By the end of the course students should be able to:

- a. have the knowledge and skills to use statistical theories
- b. have skills to test statistical hypotheses about population parameters
- c. acquire knowledge and skills in ANOVA
- d. acquire competence in correlation Theory
- e. present statistical results and interpret them in a language accessible to a non-statistician

Prerequisite

- The students should have taken and passed the following courses:
 - 1 Mathematics I and II
 - 2 Agricultural Economics I and II
 - 3 Statistics for Economists I
 - 4 Computer Applications

TOPICS OF STUDY

Topic 1: Statistical theories

- Central limit theorem, markov inequality, cauchy-shwarz theorems
- Chebyshev's theorem and Empirical Rules
- Point estimators and their properties
- Interval estimators

Topic 2: Probability Distributions

- ① Discrete Probability Distributions eg.
 - Binomial distribution
 - Trinomial distribution
 - Multinomial Distribution
 - Hypergeometric distribution

② Continuous Probability Distributions

- Normal distribution-distribution
- Exponential distribution
- Gamma and beta distribution

Topic 3: Analysis of Variance (ANOVA)

- ANOVA assumptions (Comparisons of Means vs Regression)
- Analysis of variance test: One-way classification
- ANOVA test: Two-way classifications

Topic 4: Correlations - Definition of correlation

- Types of Correlations
- Use of Correlations with real data
- Determination of validity and reliability of correlations

Topic 5: Introduction to non-parametric statistics

- Non-parametric methods: KW, Multiple Comparisons, Spearman Correlations and Point bi-serial methods
- Advantages and disadvantages of non-parametric methods
- Goodness of fit test for equal and unequal expected frequencies
- Contingency table analysis (Phi-coefficient vs Chi-square analysis)

Reference Textbooks

- ① Hogg, R.V., McKean, J.W. and Craig, A.T. (2005) Introduction to Mathematical Statistics. Pearson/Prentice Hall, Upper Saddle River, NJ, 64.
- ① Edris, A. K. (2013). Pearls of Applied Statistics. i-Publishers, Canada.
- ② Anderson, D. R., Sweeney, D.J, Williams, T.A (2002) Essentials of Statistics for Business and Economics. 2nd Edition. South Western College Publishing.
- ③ Downing, D., Clark, J. (1997) Business Statistics. 3rd Edition. New York: Barron's Education Series, inc.
- ④ Ling, D.A., Mason, R.D, and Marchal, W. G. (2000). Basic Statistics for Business and Economics. 3rd Edition. New York: Irwin McGraw-Hill. ISBN 0-07-366062-0 (Student edition)

METHOD OF ASSESSMENT

- ① Course Work: 40 %
- ② End of Course examinations: 60 %